Onboarding Logbook

Project -

Secondary Flight Storage

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Week 1 -

18th of March 2026

Tasks Completed :

- Selected onboarding project
- Created logbook design

Goals for Next Time:

- Learn basics of Altium designer
- Select a microcontroller
- Select a storage technology/module

Figure 1: Logbook Design

Week 2 -

25th of March 2026

Tasks Completed :

- Created git repo for my work
- Began creating research document
- Defined requirements
- Researched different flash storage topologies (NAND & NOR)
- Selected an IC to be used for the project (Infineon S25HL512TDPNHI010)
- Chose a microcontroller (ESP32-S3-WROOM-1U)

Goals for Next Time:

- Read important info from datasheets
- Begin PCB design
- Learn how to use git with altium for version control

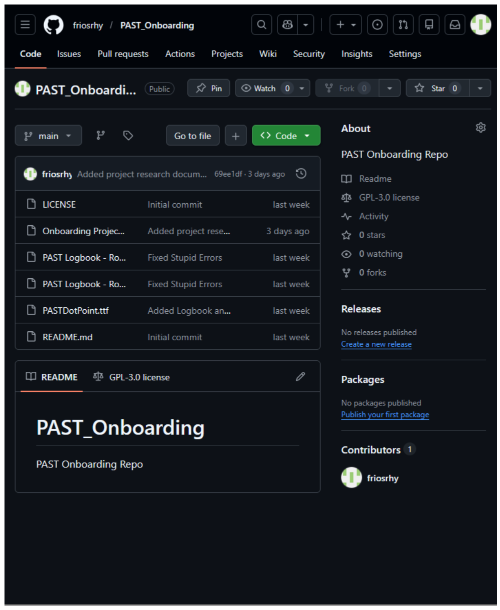


Figure 2: Github Repo

Week 3 -

1st of April 2026

Tasks Completed :

- Finished initial schematic of project
- Started research on PCB layout best practices and how to ensure signal integrity.

Goals for Next Time:

- Make changes to schematic based on feedback
- Start learning how to do SPICE and electrical checks in altium
- Start to design PCB

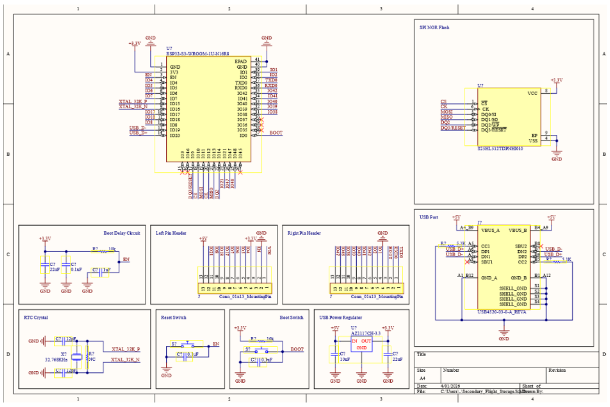


Figure 2: Initial schematic

Week 4 -

8th of April 2026

Tasks Completed :

- Reachitected schematic using recommendations and feedback on my first design
- Started using intlibs properly and having footprints associated with all of my symbols
- Completed preliminary PCB layout in KiCAD to get a basic idea of layout and confirm ideas

Goals for Next Time:

- Begin PCB layout
- Start to compile resources
- Start to design case

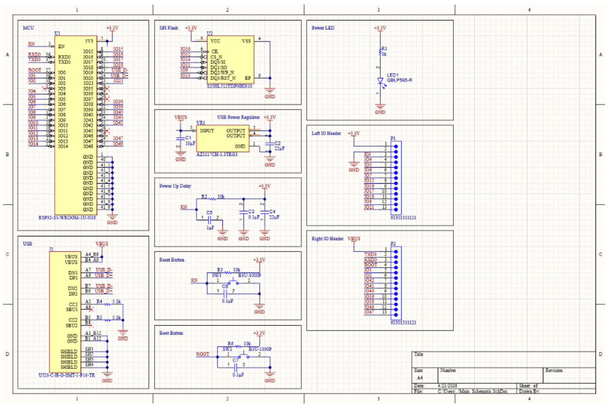


Figure 3: Reachitectured schematic

Figure 4: Basic PCB Layout In KiCAD

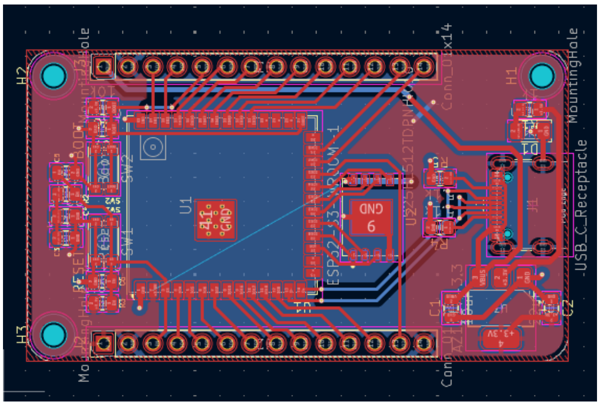
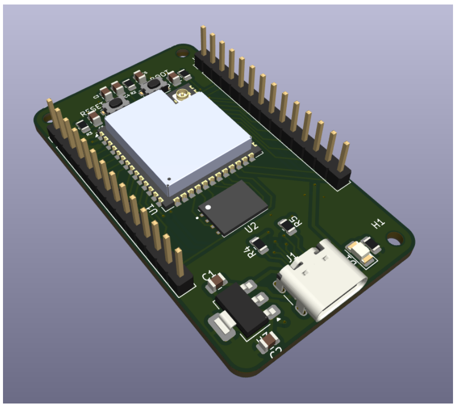


Figure 5: 3D Model of PCB



Week 5 -

15th of April 2026

Tasks Completed :

- Made basic PCB design and board outline
- Added 3d models to some footprints
- Researched trace routing best practices for USB and SPI data lines.

Goals for Next Time:

- Make final PCB layout
- Get feedback on design
- Make 3D Model and start case design

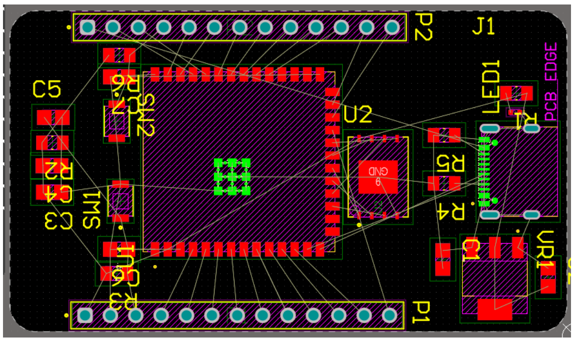


Figure 6: PCB Design in Altium